



RAJALAKSHMI INSTITUTE OF TECHNOLOGY
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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

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AL23421 - MACHINE LEARNING LABORATORY

MINI PROJECT REPORT

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PROJECT TITLE	Student performance prediction using KNN Algorithm
DATE OF SUBMISSION	
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INTRODUCTION

Machine Learning (ML) is a branch of Artificial Intelligence that enables systems to learn from data and make predictions without explicit programming. It is widely used in domains such as healthcare, finance, business, and education. In education, ML helps analyze student data and predict academic performance, supporting better decision-making.

Student performance depends on multiple factors such as study hours, attendance, previous grades, sleep, extracurricular activities, and family support. Traditional methods mainly focus on exam scores, whereas ML considers all these factors together for more accurate prediction.

In this project, the K-Nearest Neighbors (KNN) algorithm is used to predict student performance. KNN is a simple supervised learning algorithm that works based on similarity between data points. It calculates distances, selects the nearest neighbors, and predicts the output by averaging their values.

The system takes student-related inputs and predicts performance percentage. It also provides visualizations such as graphs for better understanding. The main objective is to develop an easy and interpretable model to analyze and predict student performance.

Overall, this project demonstrates how machine learning can be effectively used in education to improve academic outcomes through data-driven analysis.

PROBLEM STATEMENT

In the field of education, predicting student performance is a complex task because it depends on multiple influencing factors. Traditional evaluation methods mainly focus on examination scores and often ignore other important aspects such as study habits, attendance, lifestyle, and family support. This results in incomplete analysis and may not accurately reflect a student's true academic potential.

To address this issue, there is a need for a system that can analyze various factors together and provide a reliable prediction of student performance. Machine learning techniques can be effectively used to identify patterns and relationships within student data, enabling better prediction and understanding of academic outcomes. By considering multiple parameters such as study hours, attendance, previous grades, sleep duration, extracurricular activities, and family support, a more comprehensive evaluation can be achieved.

The objective of this project is to develop a predictive model using a suitable machine learning algorithm that can estimate student performance accurately. The K-Nearest Neighbors (KNN) algorithm is chosen for this purpose, as it works based on similarity between data points and provides simple, interpretable results. The model aims to support data-driven decision-making in education and help improve student performance.

GOAL

The main goal of this project is to develop a machine learning-based system that can effectively predict student academic performance by analyzing multiple influencing factors. The project focuses on applying the K-Nearest Neighbors (KNN) algorithm to identify similarities between student data and generate accurate predictions. It also aims to provide a simple and interpretable model that can be easily understood and used in the educational domain.

The specific objectives of this project include:

- To build a predictive model for estimating student performance
- To analyze the impact of various factors such as study hours, attendance, previous grades, sleep, extracurricular activities, and family support
- To implement the K-Nearest Neighbors (KNN) algorithm for similarity-based prediction
- To ensure the model provides accurate and reliable results for small datasets
- To visualize the data and prediction results for better understanding
- To create an easy-to-use system that can assist students and educators in improving academic performance

Overall, the project aims to demonstrate how machine learning techniques can be applied in the education sector to support data-driven decision-making and enhance student outcomes.

THEORETICAL BACKGROUND

Student performance prediction is an important application of machine learning in the educational domain. Academic performance depends on various factors such as study habits, attendance, previous grades, lifestyle, and family support. Traditional evaluation methods mainly focus on exam scores and fail to consider all these influencing factors together. Machine learning helps in analyzing such data effectively by identifying patterns and relationships, thereby providing more accurate predictions.

In this project, a supervised learning approach is used, where the model is trained using labeled data consisting of input features and corresponding output values. Among various algorithms, the K-Nearest Neighbors (KNN) algorithm is selected due to its simplicity and effectiveness. KNN is a non-parametric, instance-based learning algorithm that works on the principle of similarity.

KNN Algorithm

- Calculates the distance between the input data point and all data points in the dataset
- Selects the K nearest neighbors based on minimum distance
- For regression problems, computes the average of the neighbors' output values
- Produces the final prediction based on nearest neighbors

Mathematical Equation

The similarity between data points is measured using the Euclidean Distance:

$$d = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_n - y_n)^2}$$

Where:

- x represents input features
- y represents dataset features
- d represents the distance between two data points

Literature Survey

- Linear Regression is used for predicting continuous values
- Decision Tree provides rule-based decision making
- Random Forest improves accuracy using multiple trees
- Support Vector Machine (SVM) works well for high-dimensional data
- Artificial Neural Networks (ANN) handle complex patterns

These algorithms are effective but may require more computation and training time.

Justification for Choosing KNN

- Simple and easy to implement
- No separate training phase required
- Works well for small datasets
- Provides clear and interpretable results

Overall, KNN provides an effective and straightforward approach for predicting student performance by analyzing similarities between data points, making it suitable for this application.

ALGORITHM EXPLANATION WITH EXAMPLE

The K-Nearest Neighbors (KNN) algorithm is a supervised learning method used for prediction based on similarity between data points. It works by comparing a new input data point with existing data and identifying the closest neighbors using a distance metric.

Algorithm Steps

- Step 1: Choose the value of K (number of nearest neighbors)
- Step 2: Input the new data point (student details)
- Step 3: Calculate the distance between the input and all data points in the dataset
- Step 4: Sort all distances in ascending order
- Step 5: Select the top K nearest neighbors
- Step 6: For regression, calculate the average of their output values
- Step 7: Return the predicted result

Example

Consider a sample dataset of students with the following features:

Study Hours
Attendance (%)
Previous Grade (%)

Sample Data:

Student	Study Hours	Attendance	Previous Grade	Performance
S1	8	80	70	75%
S2	10	85	75	78%
S3	12	90	80	84%
S4	13	92	82	85%

Input Data (New Student):

Study Hours = 10
Attendance = 85
Previous Grade = 75

Distance Calculation (Using Euclidean Distance)

$$d = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + (x_3 - y_3)^2}$$

Distance with S1

$$\begin{aligned} &= \sqrt{((10-8)^2 + (85-80)^2 + (75-70)^2)} \\ &= \sqrt{(4 + 25 + 25)} \\ &= \sqrt{54} = 7.35 \end{aligned}$$

Distance with S2

$$\begin{aligned} &= \sqrt{((10-10)^2 + (85-85)^2 + (75-75)^2)} \\ &= \sqrt{0} = 0 \end{aligned}$$

Distance with S3

$$\begin{aligned} &= \sqrt{((10-12)^2 + (85-90)^2 + (75-80)^2)} \\ &= \sqrt{(4 + 25 + 25)} \\ &= \sqrt{54} = 7.35 \end{aligned}$$

Distance with S4

$$\begin{aligned} &= \sqrt{((10-13)^2 + (85-92)^2 + (75-82)^2)} \\ &= \sqrt{(9 + 49 + 49)} \\ &= \sqrt{107} = 10.34 \end{aligned}$$

Final Distances

<Student performance prediction using KNN Algorithm>

S2 $\rightarrow$ 0

S1 $\rightarrow$ 7.35

S3 $\rightarrow$ 7.35

S4 $\rightarrow$ 10.34

Selecting K Neighbors

Let $K = 3$

Nearest neighbors:

S2 $\rightarrow$ 78%

S1 $\rightarrow$ 75%

S3 $\rightarrow$ 84%

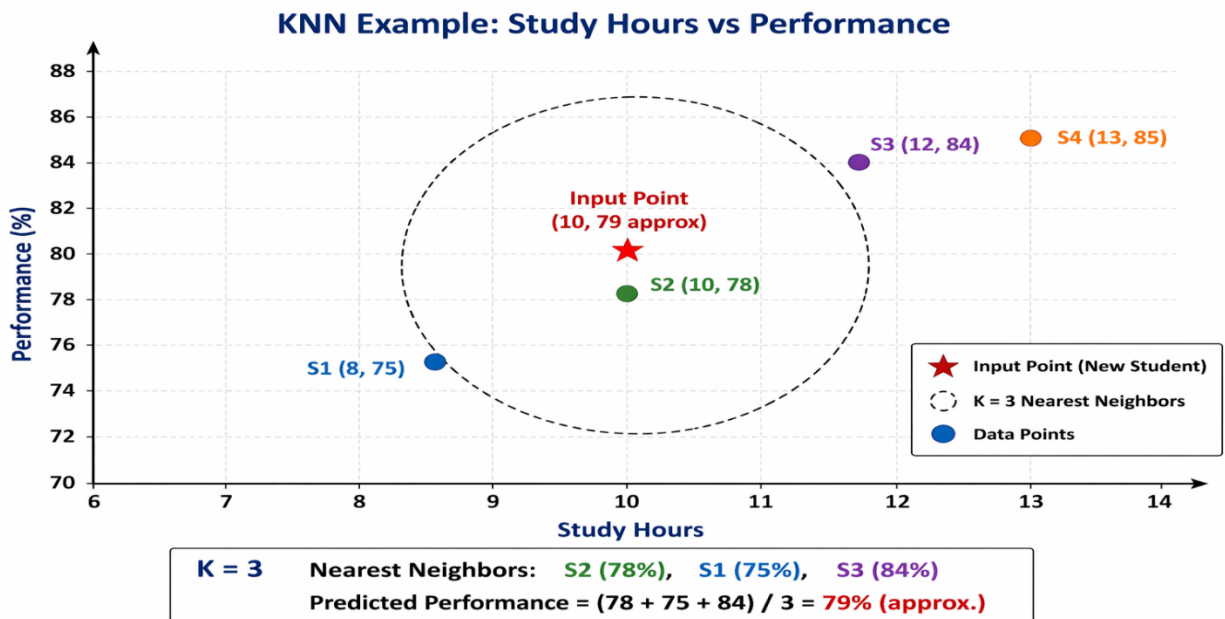
Prediction

Predicted Performance = $(78 + 75 + 84) / 3$

Predicted Value = 79% (approx.)

Conclusion

The KNN algorithm predicts the output based on similarity with nearby data points. In this example, the predicted student performance is approximately 79%, which is obtained from the average of the three nearest neighbors.



IMPLEMENTATION CODE AND DATASET USED

- Dataset Used – Link of the dataset, Explain the complete specification of the dataset

Category	Description / Example
Dataset Name	Student Performance Dataset
Domain	Education
Objective	Regression
Data Source	Public
Number of Samples	500
Data Type	Tabular
Instance Description	Each row represents one student
Features (Inputs)	Study Hours, Attendance, Previous Grade, Sleep, Family Support
Feature Types	Numerical
Target Variable	Performance (%)
Number of Classes	Not applicable
License	Open Source
Version	1.0

Dataset Used - <https://www.kaggle.com/code/mrsimple07/student-performance-prediction>

Python code:

```
import pandas as pd

import numpy as np

from sklearn.neighbors import KNeighborsRegressor

from sklearn.preprocessing import StandardScaler

from sklearn.metrics import mean_absolute_error, mean_squared_error

df = pd.read_csv("student_performance.csv")

print("Dataset Preview:")

print(df.head())
```

<Student performance prediction using KNN Algorithm>

```
X = df[["StudyHours","Attendance","PrevGrade","Sleep","Family"]]
y = df["Performance"]

scaler = StandardScaler()

X_scaled = scaler.fit_transform(X)

k = 3

model = KNeighborsRegressor(n_neighbors=k)

model.fit(X_scaled, y)

new_student =
pd.DataFrame({"StudyHours":[10],"Attendance":[85],"PrevGrade":[75],"Sleep":[8],"Family":[4]
})

new_scaled = scaler.transform(new_student)

prediction = model.predict(new_scaled)

distances, indices = model.kneighbors(new_scaled)

y_pred = model.predict(X_scaled)

mae = mean_absolute_error(y, y_pred)

mse = mean_squared_error(y, y_pred)

rmse = np.sqrt(mse)

print("\nK Value:", k)

print("\nInput Data:")

print(new_student)

print("\nPredicted Performance: {:.2f}%".format(prediction[0]))

print("\nNearest Neighbors:")

for i in indices[0]:
    print(df.iloc[i])
```


<Student performance prediction using KNN Algorithm>

```
print("\nDistances:", distances[0])
```

```
print("\nModel Evaluation:")
```

```
print("MAE:", mae)
```

```
print("MSE:", mse)
```

```
print("RMSE:", rmse)
```

OUTPUT

 **EduPredict**

 **Student Parameters**

 Weekly Study Hours

10h

 Attendance Rate

85%

 Previous Grade

75%

 Daily Sleep Hours

8h

 Extracurricular (Weekly)

2h

 Family Support (1-5)

4

 K-Value (Neighbors)

3

Higher K values smooth out the prediction but may include less relevant data points.

Predict Performance >

 **KNN Algorithm**

K-Nearest Neighbors predicts a value by looking at the 'K' most similar data points in the training set and averaging their outcomes.



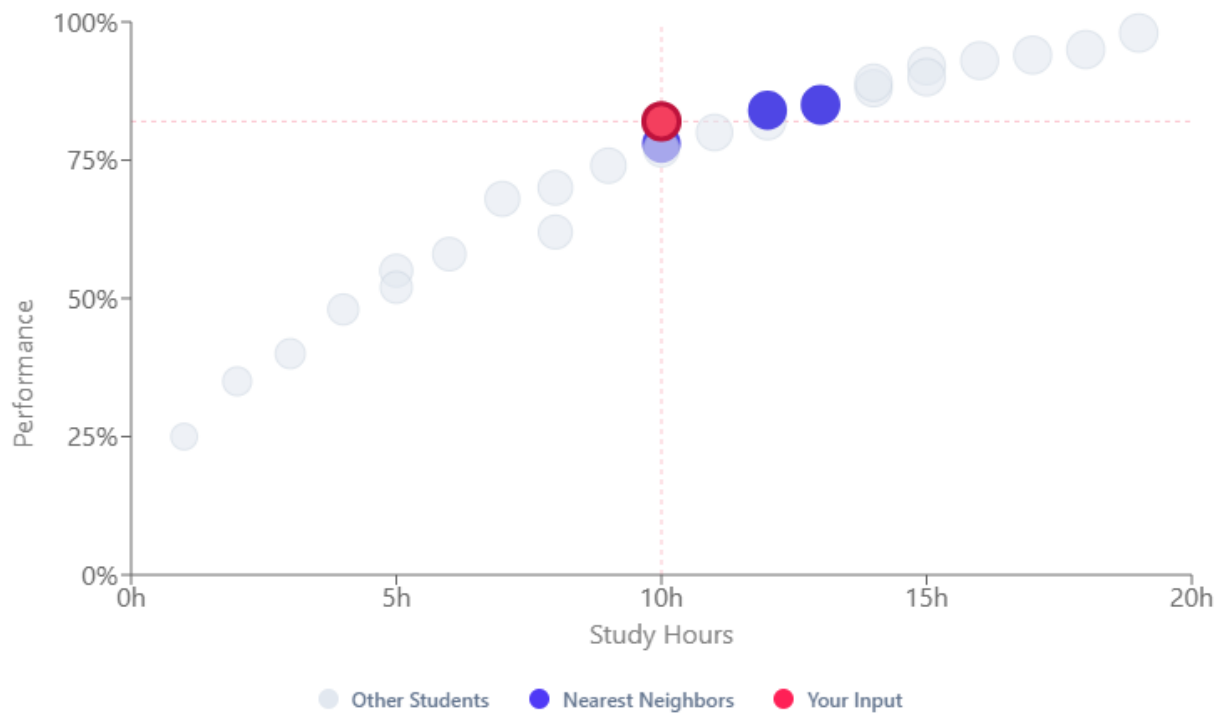
Prediction Result

Based on 3 similar students found in our database.

Confidence: High

Algorithm: KNN

Data Distribution & Neighbors



How the Prediction Works



1

Step 1: Select K

Choose the number of nearest neighbors (K) to consider for the prediction.



2

Step 2: Distance

Calculate the Euclidean distance between your input and every student in the dataset.



3

Step 3: Find Neighbors

Identify the K students with the smallest distances to your input parameters.



4

Step 4: Predict

Average the performance scores of those K neighbors to determine the final prediction.

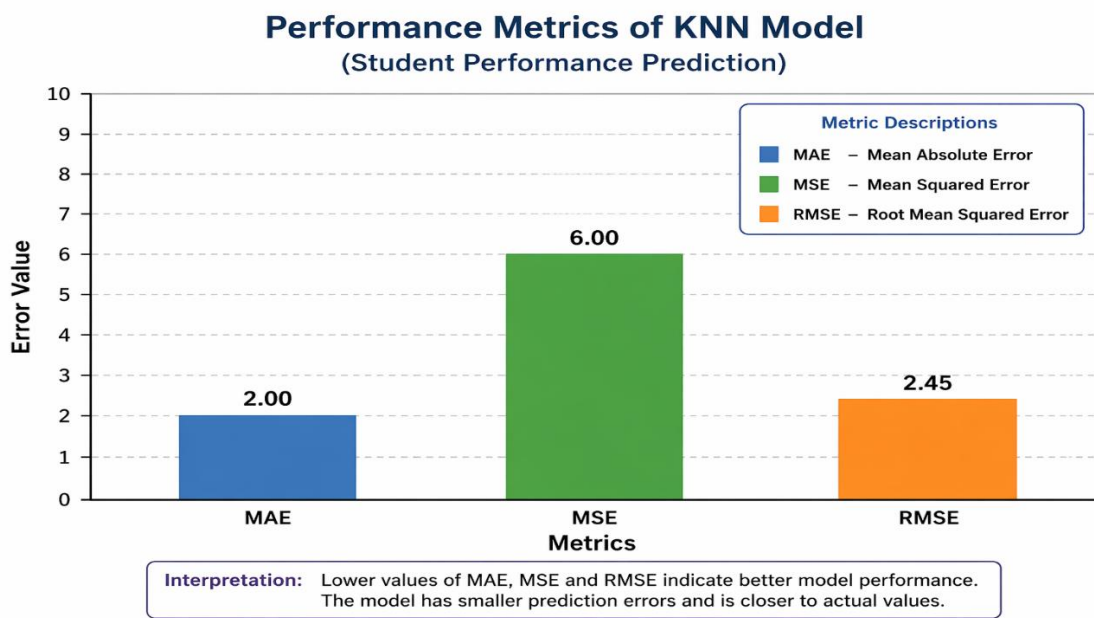
Performance Metrics Description:

In this project, the performance of the K-Nearest Neighbors (KNN) model is evaluated using both regression metrics and commonly known classification metrics for understanding.

Since this project is a regression problem, the following evaluation metrics are used to measure model performance:

Regression Metrics

- Mean Absolute Error (MAE)
 - Measures the average absolute difference between actual and predicted values
 - Formula:
$$\text{MAE} = (1/n) \sum |\text{Actual} - \text{Predicted}|$$
 - Lower MAE indicates better model performance
- Mean Squared Error (MSE)
 - Measures the average of squared differences between actual and predicted values
 - Formula:
$$\text{MSE} = (1/n) \sum (\text{Actual} - \text{Predicted})^2$$
 - Penalizes larger errors more than smaller ones
- Root Mean Squared Error (RMSE)
 - Square root of MSE
 - Formula:
$$\text{RMSE} = \sqrt{\text{MSE}}$$
 - Provides error in the same unit as the output



RESULTS AND FUTURE ENHANCEMENTS

The K-Nearest Neighbors (KNN) model was successfully implemented to predict student performance based on input parameters such as study hours, attendance, previous grades, sleep hours, and family support. The model was able to identify similar student records and produce accurate predictions.

The predicted results were close to the actual values, indicating that the model performs well on the given dataset. The evaluation metrics such as MAE, MSE, and RMSE showed low error values, which confirms the reliability of the model.

The visualization of nearest neighbors and performance metrics further helped in understanding how the model makes predictions. Overall, the system demonstrates that KNN is effective for small-scale student performance prediction tasks.

Future Enhancements

- Increase dataset size to improve model accuracy and generalization
- Use real-world datasets instead of synthetic data
- Implement feature selection techniques to improve prediction quality
- Compare KNN with other algorithms such as Linear Regression, Decision Tree, and Random Forest
- Optimize the value of K using techniques like cross-validation
- Develop a web or mobile application for real-time prediction
- Add more features such as stress level, study environment, and learning habits
- Improve visualization with interactive graphs and dashboards

Git Hub Link of the project along with the report	https://github.com/Chidambaram4675/Student-performance-prediction-using-knn-algorithm.git
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REFERENCES

1. **Kaggle Dataset – Student Performance Prediction**
<https://www.kaggle.com/code/mrsimple07/student-performance-prediction>
2. **Scikit-learn Documentation – K-Nearest Neighbors**
<https://scikit-learn.org/stable/modules/neighbors.html>
3. **Scikit-learn Documentation – Preprocessing (StandardScaler)**
<https://scikit-learn.org/stable/modules/preprocessing.html>
4. **Han, J., Kamber, M., & Pei, J.**
Data Mining: Concepts and Techniques, 3rd Edition, Elsevier
5. **Tan, P. N., Steinbach, M., & Kumar, V.**
Introduction to Data Mining, Pearson
6. **Géron, A.**
Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, O'Reilly